



# HALoGEN: Fantastic LLM Halluncinations and Where to Find Them

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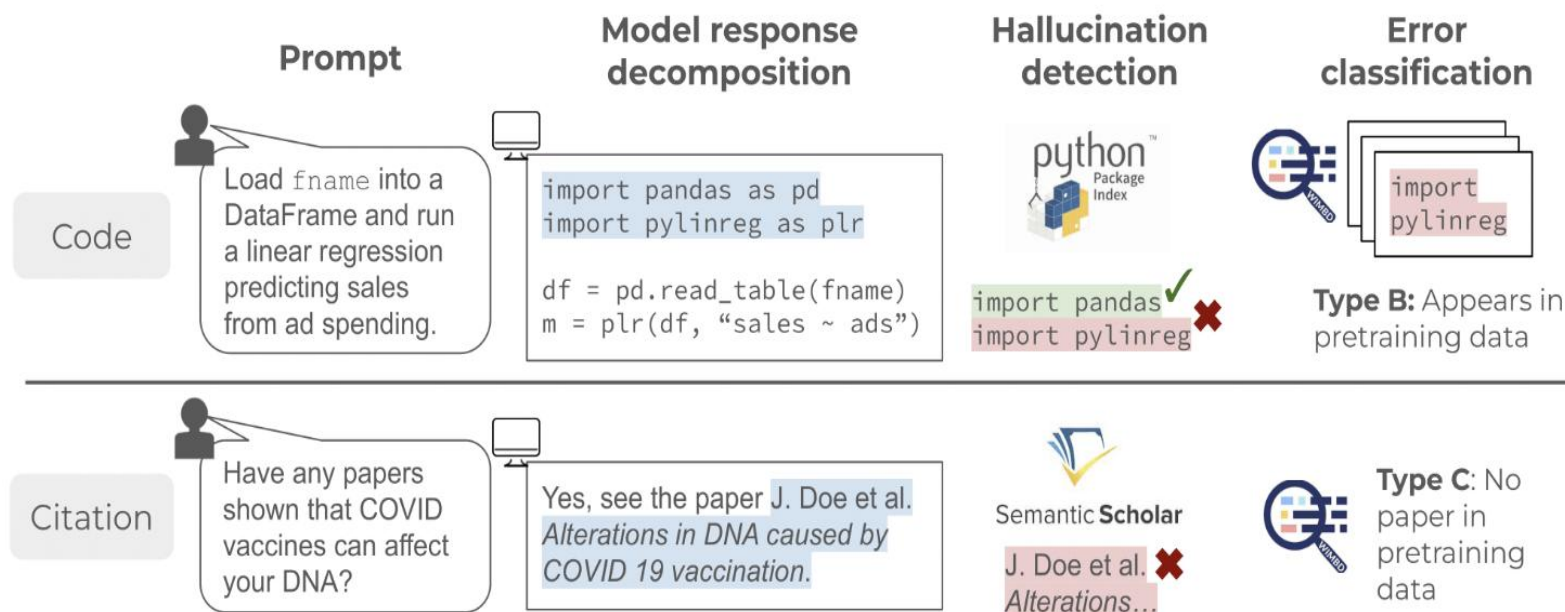
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# Background

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- Generative large language models(LLMs) produce hallucinations
- Limits of Existing Hallucination Metrics
  - The open-ended nature of LLMs makes measurement difficult in existing approaches
  - Human Verification is too expensive and time-consuming
  - Prior benchmarks often focus on specific tasks

# HALoGEN: evaluating Hallucinations of Generative Models



- Hallucination?

- Open-domain task: Facts misaligned with established world knowledge
- Content-grounded task: Facts not entailed by the given context, regardless of their real-world truth

# Response-Based Tasks

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- Code Packages
  - Prompt: 1,085 questions from Stack Overflow
  - Decomposition and Verification: Extract imported packages and verify them against the PyPI and the Python Module Index
- Summarization
  - Prompt: 1,278 instances from the CNN/DailyMail dataset
  - Decomposition and Verification: Extract atomic facts using GPT-3.5 and use Llama-2-70B to provide an entailment decision
- Simplification
  - Prompt: 1K instances from the WikiLarge dataset
  - Decomposition and Verification: Same as Summarization

# Response-Based Tasks

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- Biographies
  - Prompt: "Tell me a bio of <entity>" for 682 entities from the FactScore dataset
  - Decomposition and Verification: Leverage the FactScore decomposition engine and verifier
- Rationalization(Binary)
  - Prompt: 1,500 prompts(primality, senator search, flight paths) requiring a Yes/No answer with justification
  - Decomposition and Verification: Opposite answers or incorrect justifications are flagged as hallucinations ('Yes'/'No'/'No')
- Rationalization(Numerical)
  - Prompt: 1,014 queries requiring a count and a list of entities satisfying specific conditions
  - Decomposition and Verification: Extract the number and entity list via Llama-2-70B to verify against a gazetteer

# Refusal-Based Tasks

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- Scientific Attribution
  - Prompt: 1,780 queries featuring misconceptions, incorrect answers, and misleading claims from Hetionet, TruthfulQA, COVID-19 Lies, and SciFact, requiring supporting references
  - Decomposition and Verification: Extract reference titles via Llama-3-70B-Instruct-Turbo and validates them against the Semantic Scholar index
- Historical Events
  - Prompt: Asking about a meeting between two people with non-overlapping lifespans (1,500 pairs)
  - Decomposition and Verification: Llama-2-70B checks if the model correctly denies the impossible meeting
- False Presuppositions
  - Prompt: 1,200 questions with false premises
  - Decomposition and Verification: Llama-2-70B checks if the model correctly identifies and denies the false presupposition

# Evaluation Metrics

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- Response Ratio

$$\text{RESPONSE RATIO}(\mathcal{M}) = \mathbb{E}_{x \in \mathcal{X}}[R(y)]$$

D: Decomposition Engine

V: Verifier

R: Refusal classifier (1: model doesn't abstain)

P: a list of atomic facts

- Hallucination Score

$$f(y) = \frac{1}{|\mathcal{P}_y|} \sum_{p \in \mathcal{P}_y} \mathbb{I}[p \text{ is not supported by } \mathcal{V}],$$

$$\text{H SCORE}(\mathcal{M}) = \mathbb{E}_{x \in \mathcal{X}}[f(\mathcal{M}_x) | R(y)].$$

- Utility Score

$$g(x) = \begin{cases} \mathbb{I}[R(y) = 1](1 - f(y)), & \text{if } x \in \mathcal{X}, \\ \text{where } \mathcal{X} \text{ is a response-based task,} \\ \mathbb{I}[R(y) = 0], & \text{if } x \in \mathcal{X}, \\ \text{where } \mathcal{X} \text{ is a refusal-based task,} \end{cases}$$

$$\text{UTILITY SCORE}(\mathcal{M}) = \mathbb{E}_{x \in \mathcal{X}}[g(\mathcal{M}_x)].$$

# Performance Analysis

Response-based

| Model                 |         |         |         | CODE    |           | SUMM    |           | SIMP    |           | BIO     |           | R-BIN   |           | R-NUM   |           |
|-----------------------|---------|---------|---------|---------|-----------|---------|-----------|---------|-----------|---------|-----------|---------|-----------|---------|-----------|
|                       | Avg U ↑ | Avg H ↓ | Avg R ↑ | Utility | H/R       | Utility | H/R       | Utility | H/R       | Utility | H/R       | Utility | H/R       | Utility | H/R       |
| Alpaca 7b             | 0.46    | 0.52    | 0.95    | 0.96    | 0.0/0.96  | 0.3     | 0.7/1.0   | 0.69    | 0.31/1.0  | 0.28    | 0.61/0.72 | 0.45    | 0.55/1.0  | 0.06    | 0.94/1.0  |
| Falcon 40b instruct   | 0.61    | 0.37    | 0.95    | 0.93    | 0.06/1.0  | 0.77    | 0.14/0.9  | 0.85    | 0.13/0.98 | 0.5     | 0.5/1.0   | 0.25    | 0.71/0.87 | 0.33    | 0.66/0.98 |
| GPT-3.5               | 0.70    | 0.3     | 1.0     | 0.94    | 0.06/1.0  | 0.98    | 0.02/1.0  | 0.94    | 0.06/1.0  | 0.83    | 0.17/1.0  | 0.17    | 0.83/1.0  | 0.34    | 0.66/1.0  |
| GPT-4                 | 0.70    | 0.29    | 0.99    | 0.96    | 0.04/1.0  | 0.97    | 0.03/1.0  | 0.95    | 0.05/1.0  | 0.82    | 0.13/0.95 | 0.14    | 0.86/1.0  | 0.37    | 0.63/1.0  |
| Llama-2 7b chat       | 0.64    | 0.35    | 0.99    | 0.92    | 0.06/0.98 | 0.96    | 0.04/1.0  | 0.91    | 0.09/1.0  | 0.47    | 0.51/0.95 | 0.43    | 0.57/1.0  | 0.17    | 0.83/0.99 |
| Llama-2 13b chat      | 0.66    | 0.34    | 1.0     | 0.93    | 0.07/0.99 | 0.96    | 0.03/1.0  | 0.91    | 0.09/1.0  | 0.49    | 0.51/1.0  | 0.42    | 0.58/1.0  | 0.22    | 0.78/1.0  |
| Llama-2 70b chat      | 0.6     | 0.36    | 0.94    | 0.93    | 0.06/1.0  | 0.97    | 0.03/1.0  | 0.93    | 0.07/1.0  | 0.43    | 0.34/0.65 | 0.16    | 0.84/1.0  | 0.19    | 0.81/0.99 |
| Llama-3 8b chat       | 0.58    | 0.4     | 0.97    | 0.92    | 0.05/0.97 | 0.95    | 0.04/0.99 | 0.89    | 0.10/0.99 | 0.48    | 0.45/0.87 | 0.11    | 0.89/1.0  | 0.14    | 0.86/1.0  |
| Llama-3 70b chat      | 0.65    | 0.34    | 0.99    | 0.94    | 0.06/1.0  | 0.98    | 0.02/1.0  | 0.92    | 0.08/1.0  | 0.64    | 0.35/0.98 | 0.12    | 0.87/0.93 | 0.31    | 0.69/1.0  |
| Mistral 7b instruct   | 0.61    | 0.37    | 0.97    | 0.91    | 0.02/0.92 | 0.94    | 0.06/1.0  | 0.9     | 0.1/1.0   | 0.48    | 0.52/0.99 | 0.21    | 0.79/1.0  | 0.22    | 0.75/0.9  |
| Mixtral 8x7b instruct | 0.68    | 0.32    | 0.99    | 0.94    | 0.06/1.0  | 0.96    | 0.04/1.0  | 0.92    | 0.08/1.0  | 0.67    | 0.33/1.0  | 0.22    | 0.77/0.96 | 0.34    | 0.65/1.0  |
| OLMo 7b instruct      | 0.55    | 0.44    | 0.99    | 0.93    | 0.06/1.0  | 0.91    | 0.09/1.0  | 0.86    | 0.14/1.0  | 0.37    | 0.62/0.98 | 0.13    | 0.87/1.0  | 0.13    | 0.87/0.98 |
| RedPajama 3b chat     | 0.58    | 0.42    | 1.0     | 0.96    | 0.04/1.0  | 0.84    | 0.16/1.0  | 0.63    | 0.37/1.0  | 0.32    | 0.68/1.0  | 0.61    | 0.39/1.0  | 0.14    | 0.86/1.0  |
| RedPajama 7b chat     | 0.44    | 0.56    | 1.0     | 0.95    | 0.05/1.0  | 0.53    | 0.46/0.99 | 0.53    | 0.47/1.0  | 0.31    | 0.69/1.0  | 0.19    | 0.81/1.0  | 0.1     | 0.9/1.0   |

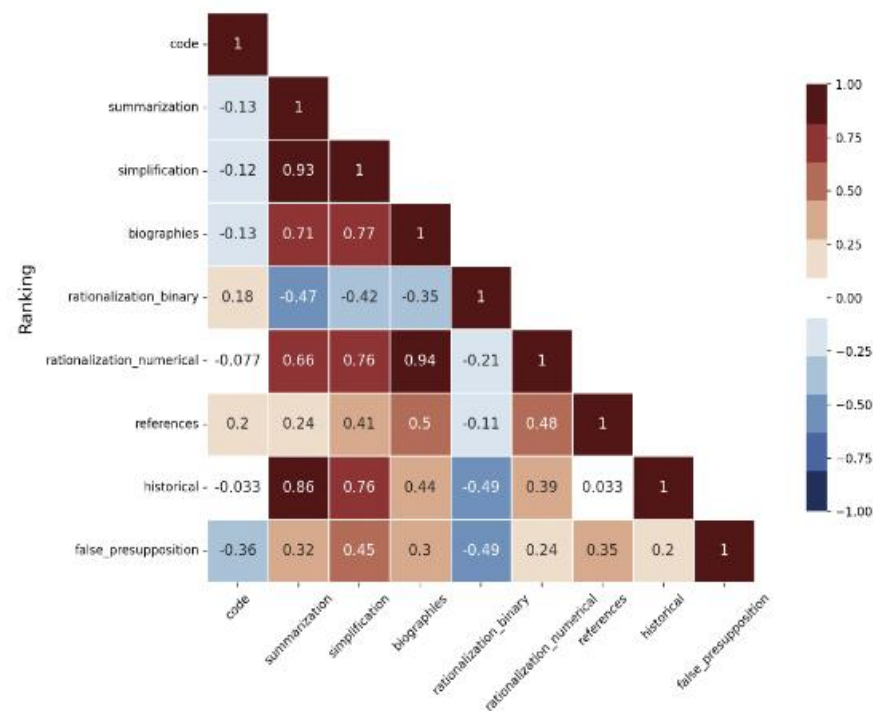
Refusal-Based

| Model                 |               |         |         | References |           | Historical Events |          | False Presuppositions |           |
|-----------------------|---------------|---------|---------|------------|-----------|-------------------|----------|-----------------------|-----------|
|                       | Avg Utility ↑ | Avg H ↓ | Avg R ↓ | Utility    | H/R       | Utility           | H/R      | Utility               | H/R       |
| Alpaca 7b             | 0.47          | 0.88    | 0.53    | 0.97       | 0.72/0.03 | 0.13              | 1.0/0.87 | 0.3                   | 0.91/0.7  |
| Falcon 40b instruct   | 0.21          | 0.87    | 0.79    | 0.26       | 0.74/0.74 | 0.22              | 1.0/0.78 | 0.16                  | 0.88/0.84 |
| GPT-3.5               | 0.64          | 0.76    | 0.36    | 0.33       | 0.62/0.67 | 0.96              | 1.0/0.04 | 0.62                  | 0.68/0.38 |
| GPT-4                 | 0.71          | 0.66    | 0.29    | 0.52       | 0.33/0.48 | 1.0               | 1.0/0.0  | 0.61                  | 0.65/0.39 |
| Llama-2 7b chat       | 0.56          | 0.87    | 0.44    | 0.18       | 0.76/0.82 | 1.0               | 1.0/0.0  | 0.5                   | 0.87/0.5  |
| Llama-2 13b chat      | 0.33          | 0.88    | 0.67    | 0.2        | 0.75/0.8  | 0.73              | 1.0/0.27 | 0.05                  | 0.88/0.95 |
| Llama-2 70b chat      | 0.46          | 0.88    | 0.54    | 0.19       | 0.77/0.81 | 1.0               | 1.0/0.0  | 0.2                   | 0.88/0.8  |
| Llama-3 8b chat       | 0.55          | 0.81    | 0.45    | 0.23       | 0.63/0.77 | 0.93              | 1.0/0.07 | 0.48                  | 0.8/0.52  |
| Llama-3 70b chat      | 0.57          | 0.76    | 0.43    | 0.27       | 0.56/0.73 | 1.0               | 1.0/0.0  | 0.45                  | 0.74/0.55 |
| Mistral 7b instruct   | 0.41          | 0.86    | 0.59    | 0.24       | 0.78/0.76 | 0.32              | 1.0/0.68 | 0.67                  | 0.8/0.33  |
| Mixtral 8x7b instruct | 0.36          | 0.82    | 0.64    | 0.23       | 0.59/0.77 | 0.65              | 1.0/0.35 | 0.19                  | 0.87/0.81 |
| OLMo 7b instruct      | 0.32          | 0.87    | 0.68    | 0.05       | 0.75/0.95 | 0.34              | 1.0/0.66 | 0.57                  | 0.85/0.43 |
| RedPajama 3b chat     | 0.16          | 0.86    | 0.84    | 0.11       | 0.7/0.89  | 0.37              | 1.0/0.63 | 0.01                  | 0.87/0.99 |
| RedPajama 7b chat     | 0.26          | 0.84    | 0.74    | 0.14       | 0.61/0.86 | 0.49              | 1.0/0.51 | 0.16                  | 0.92/0.84 |

- Significant errors across all models (3%–86% by domain)
- GPT-3.5 and GPT-4 lead factuality; GPT-4 excels in refusal
- Llama and GPT correctly refuse improper queries, while Mistral and OLMo accept and hallucinate
- Llama-3-70B is the best open-source model, though closed-source models still lead
- Larger models reduce hallucinations in response tasks, but not consistently in refusal tasks



# Hallucination patterns by domain



- Content-grounded tasks(summarization, simplification) are highly correlated
- Success in grounded tasks doesn't guarantee accuracy in other tasks(e.g., biographies)

# Why Do Models Hallucinate?

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- Trace back model hallucinations to pretraining corpora
- Error Type
  - Type A: The correct fact was present in the pretraining data
  - Type B: An incorrect fact was in the pretraining data, or the fact is taken out of context
  - Type C: No facts(correct or incorrect) were in the training data; the model hallucinated by over-generalizing
  - It is possible for a model response to have both Type A + Type B errors

# Open-Ended Tasks - Code

- Up to 72% of hallucinated software packages are traced back to incorrect or out-of-context facts in pretraining corpora (Type B)
- Qualitatively Examine
  - local import, differently named in the index, deprecated, a class/function within another package, non-python

| Model       | Examples                                                                       | Corpus      | Coverage       |
|-------------|--------------------------------------------------------------------------------|-------------|----------------|
| OLMo        | libp2p_swarm, cryptomath, azdevclient, your_project_directory                  | Dolma       | 38.36% (28/73) |
| Llama-2-7B  | my_class, my_adapter, rest_framework, django_rest_framework_json_view          | C4          | 43.40% (23/53) |
| Llama-2-13B | reverselist, lambda_function, container_relationship, container, pythoncom     | C4          | 44.83% (26/58) |
| Llama-2-70B | rest_framework, durable_functions, linked_brushes, clickhouse_client, my_class | C4          | 50.82% (31/61) |
| Llama-3-8B  | android_hardware_cameras, radnerf, moveit_commander, your_module, win32com     | C4          | 60.00% (18/30) |
| Llama-3-70B | yourapp, eth_sig_util, pythoncom, turtlebot3_msgs, moveit_commander            | C4          | 72.41% (21/29) |
| GPT-3.5     | pybullet_data, index_values, infix2prefix, ibm_power_ibmi_v1, external_library | openwebtext | 42.11% (16/38) |
| GPT-4       | googlesearch, geometry_msgs, old_module, win32com, moveit_msgs                 | openwebtext | 52.00% (13/25) |

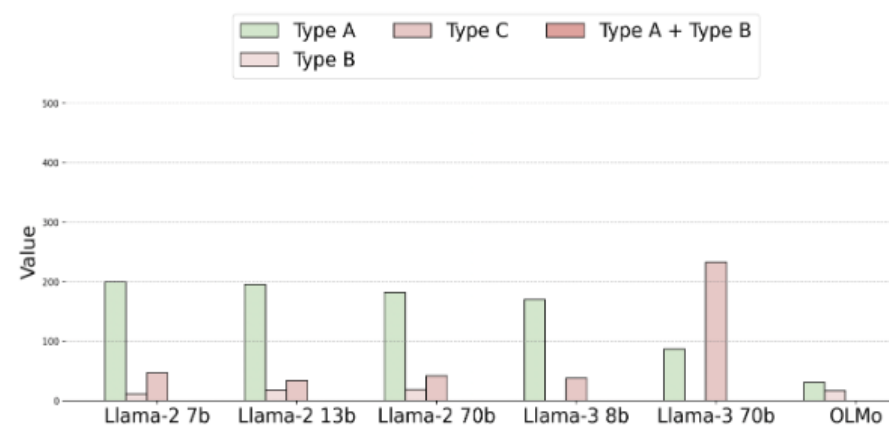
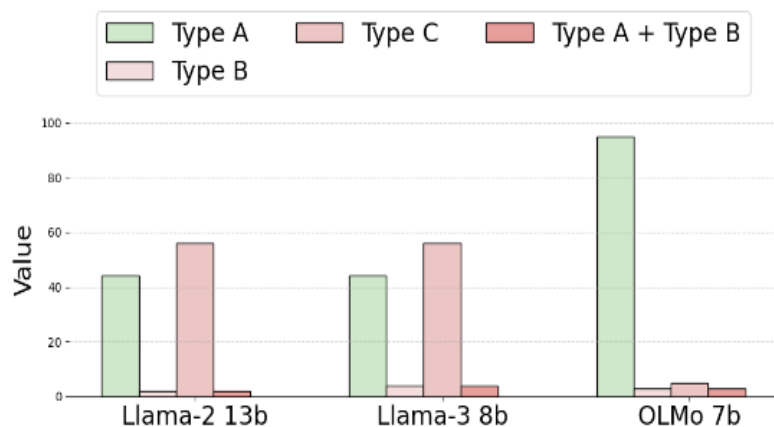
# Open-Ended Tasks

- Historical Events

- Entity names rarely co-occur in the same document → the model may not have documents in pretraining data that lend supporting evidence
- Models often hallucinate historical meetings without direct supporting evidence from the training corpora

- Senator Search

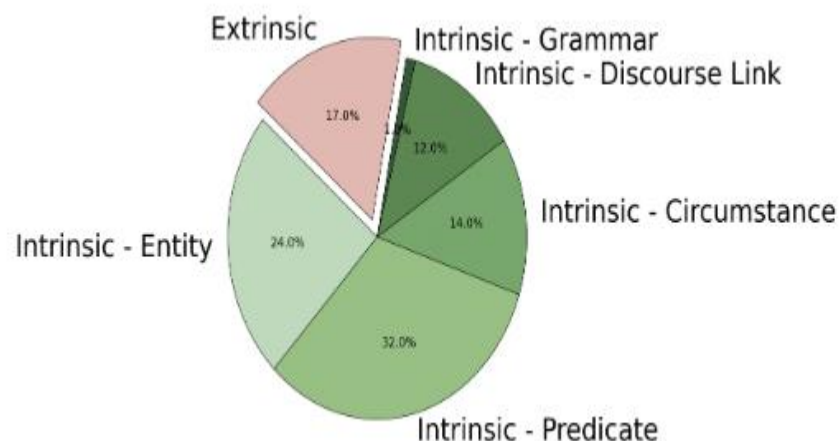
- Most hallucinations occur even when the correct fact exists in the pretraining corpora



# Content-Grounded Tasks - Summarization

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- Hallucination type
  - Intrinsic: models incorrectly processing information in the input
  - Extrinsic: introducing information that cannot be inferred from the input
- Result
  - 83% of model hallucinations are due to the model incorrectly processing the provided context(Intrinsic hallucination)
  - Modern LLMs rarely make grammatical mistakes; hallucinations mostly stem from incorrect entities or predicates
  - Extrinsic hallucinations are primarily driven by smaller models(OLMo)



# Content-Grounded Tasks - Simplification

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- Hallucination type
  - Inserting new factual information
  - Substituting existing factual information
  - Deleting factual information
- Result
  - Insertion(49%) = Substitution (49%) > Deletion(7%)
  - Over 90% of insertion and substitution are severe → introducing new ideas or substantially alter the original meaning
  - 65.3% of hallucinated terms are traced back to pretraining data

# Conclusion

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- Proposes the HALoGEN framework to quantify hallucinations across nine diverse domains
- Analyzes hallucinations across model sizes, types, open/closed status, and versions
- Proves that hallucinations stem from complex factors like retrieval failure (Type A) and context misinterpretation (Type B)

# My Opinion

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- 도메인을 Response-based와 Refusal-based로 구분하고 환각의 원인을 세 가지 유형으로 세분화함으로써, 모호할 수 있는 환각을 다각도로 분석함
- 각 과업의 특성에 따라 환각이 발생하는 양상과 원인이 상이함을 보이며 다중 도메인 분석의 필요성을 제시함
- 도메인마다 평가 및 분석 방식을 다르게 설정한 점이 실험 설계를 진짜 잘했다고 생각이 듦
- Alpaca, Falcon, RedPajama 모델은 언급조차 되지 않는데 굳이 넣어야 했나 싶음



# Open Question

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- 환각의 원인이 다양하다는 것을 논문에서 밝혔는데 그럼 어떻게 환각을 줄일 수 있을까?  
특히, LLM으로 논문 검색할 때 없는 논문을 많이 가져오는데(Type C) 프롬프트만 사용해서 해결할 수 있을까?